REPUBLIC OF TUNISIA
Ministry of Higher Education and Scientific Research
University of Carthage
National Institute of Applied Sciences and Technology

Graduation Project

To obtain the

National Engineering Diploma

In Applied Sciences and Technology

Specialty : Software Engineering

Design, Implementation, and Pipeline Integration of a Smoke Test Suite as a CI/CD Quality Gate for a Multi-Universe SaaS Platform

---

Presented by :

Ahmed SILINI

Hosted By

LittleBig Connection

Defended on the 9th of June 2026 in front of a jury composed of :

Mr. Emir DAMERJI	President of the jury
Mrs. Lilia SFAXI	Reviewer
Mrs. Hajer AKAICHI	Supervisor at the Company
Mrs. Sana HAMDI	Supervisor at INSAT

University year : 2025 / 2026

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Supervisor at the Company Mrs. Hajer AKAICHI	Supervisor at INSAT Mrs. Sana HAMDI
Date :	Date :

University year : 2025 / 2026

Dedication

Write your dedication here.

To those you wish to honour, in a few short lines.

[Your Name]

Acknowledgements

Thank the people who supported your work here: your company supervisor and mentor, your academic supervisor, your team, the jury members, and your institution. Replace the bracketed names below with your own.

I am grateful to my company supervisor, **[Supervisor Name]**, and my mentor, **[Mentor Name]**, for their guidance throughout the internship.

I thank my academic supervisor, **[Academic Supervisor Name]**, for the attention and feedback given to this project.

I also thank the members of the jury for the time invested in evaluating this work.

[Your Name]

Abstract

Write a short English abstract here: the context, the problem, what you built, and the main results, in one or two paragraphs.

Keywords: keyword one, keyword two, keyword three.

Résumé

Rédigez ici un court résumé en français : le contexte, la problématique, ce que vous avez réalisé et les principaux résultats, en un ou deux paragraphes.

Mots-clés : mot-clé un, mot-clé deux, mot-clé trois.

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List of Figures

1.1 Replace this with your own figure. Put images under `assets/`. 3

List of Tables

1.1 Replace this with your own table. 3

List of Acronyms

API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Delivery
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
UI	User Interface
URL	Uniform Resource Locator

General Introduction

Write your general introduction here. It usually sets the broad context, states the problem your project addresses, summarises your contribution, and ends with a paragraph announcing how the report is organised chapter by chapter.

Chapter 1

First Chapter Title

Summary

1.1 A Section (with figures, tables, code)
1.2 Lists and Cross-references

Introduction

Open every chapter with a short paragraph that states what the chapter covers and how it connects to the previous one. Replace this placeholder with your own introduction.

1.1 A Section

Write your prose here. Start a new paragraph by leaving a blank line. Use **bold** and *italic* sparingly for emphasis.

1.1.1 Including a Figure

Reference a figure in the text with Figure 1.1, then place the figure with the `[H]` option so it stays where you put it.



Figure 1.1: Replace this with your own figure. Put images under `assets/`.

1.1.2 Including a Table

Reference a table with Table 1.1. Use `tabularx` so the table fills the text width and wraps long cells automatically.

Column A	Column B
Row label	A cell that can hold a long sentence and wrap onto the next line without breaking the table layout.
Another row	Another description.

Table 1.1: Replace this with your own table.

1.1.3 Including a Code Listing

Reference a listing with Listing 1.1.

Listing 1.1: Replace with your own code.

```
def greet(name):
    return f"Hello, {name}!"
```

1.1.4 Citing a Reference

Cite a source with `\cite`, like this [1]. Add the entry to `preliminary pages/bibliography.bib`; it appears in the Bibliography automatically.

1.2 Lists and Cross-references

An unordered list:

- First point.
- Second point.

An ordered list:

1. First step.
2. Second step.

Refer back to earlier material by label, e.g. section 1.1.

Conclusion

Close every chapter with a short paragraph that summarises it and bridges to the next one.

Chapter 2

Second Chapter Title

Summary

2.1 First Section
2.2 Second Section

Introduction

Introduce the chapter here.

2.1 First Section

Your content here.

2.2 Second Section

Your content here.

Conclusion

Summarise the chapter and bridge to the next one.

Chapter 3

Third Chapter Title

Summary

3.1 First Section
3.2 Second Section

Introduction

Introduce the chapter here.

3.1 First Section

Your content here.

3.2 Second Section

Your content here.

Conclusion

Summarise the chapter and bridge to the next one.

Chapter 4

Fourth Chapter Title

Summary

4.1 First Section
4.2 Second Section

Introduction

Introduce the chapter here.

4.1 First Section

Your content here.

4.2 Second Section

Your content here.

Conclusion

Summarise the chapter and bridge to the next one.

Chapter 5

Fifth Chapter Title

Summary

5.1 First Section
5.2 Second Section

Introduction

Introduce the chapter here.

5.1 First Section

Your content here.

5.2 Second Section

Your content here.

Conclusion

Summarise the chapter and bridge to the next one.

General Conclusion and Perspectives

Write your general conclusion here. Restate what the project delivered, what you learned, and the limitations you ran into. Close with a *Perspectives* paragraph (or a short list) outlining concrete next steps and future improvements.

Bibliography

[1] F. Lastname, *An Example Book Title*. Publisher Name, 2024, ISBN: 978-0000000000.

Appendices

Appendix A: Title

Place supplementary material here (large tables, extra figures, configuration files). Reference an appendix from the text with Appendix A.

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تصميم وتنفيذ مجموعة اختبارات الدخان ودمجها في خط الأنابيب كبوابة جودة للتكامل المستمر والنشر المستمر (CI/CD) لمنصة برمجية كخدمة (SaaS) متعددة الأكوان

الملخص

تُشغّل LittleBig Connection منصة برمجية كخدمة (SaaS) موزعة على أربعة أكوان تطبيقية. كانت مجموعة اختبارات الانحدار الخاصة بفريق ضمان الجودة، والمكوّنة من ٩٧٦ حالة اختبار مبنية باستخدام Robot Framework وSeleniumLibrary، تتطلب أكثر من ثماني ساعات للتنفيذ مع معدل هشاشة غير مقاس. يقدّم هذا العمل مجموعة اختبارات دخان مبنية باستخدام Robot Framework وBrowser Library المعتمدة على Playwright، تعمل كبوابة جودة في المرحلة الأولى من خط أنابيب التكامل المستمر والنشر المستمر (CI/CD) على GitLab، وتمنع تنفيذ اختبارات الانحدار عند فشل أي مسار حرج. تغطي المجموعة ١٨ اختباراً مستقلاً لأهم ٢٠ مساراً عبر مجالات Auth وAdmin وClient وSupplier، وتُنجز في حوالي دقيقتين بالتنفيذ المتوازي، أي أسرع بنحو ٢.٩ مرة مقارنة بمسارات Selenium المكافئة. كما تُنشر النتائج تلقائياً على Microsoft Teams وتُتابع عبر لوحة معلومات GitLab Pages.

الكلمات المفتاحية: Playwright، Browser Library، Robot Framework، اختبارات الدخان، الاختبارات الشاملة، التكامل المستمر والنشر المستمر (CI/CD)، GitLab، بوابة الجودة، هشاشة الاختبارات، استقلالية الاختبارات، نموذج كائن الصفحة، DevOps.

Conception, implémentation et intégration dans le pipeline d'une suite de tests smoke comme barrière qualité CI/CD pour une plateforme SaaS multi-univers

Résumé

LittleBig Connection exploite une plateforme SaaS sur quatre univers applicatifs. La suite de régression QA (976 tests sur Robot Framework et SeleniumLibrary) dépassait huit heures d'exécution avec un flakiness non mesuré. Le travail introduit une suite smoke basée sur Robot Framework et Browser Library/Playwright, exécutée en première étape du pipeline GitLab CI/CD pour bloquer la régression dès qu'un parcours critique échoue. Dix-huit tests indépendants couvrent les vingt parcours prioritaires des univers Auth, Admin, Client et Fournisseur, s'exécutent en environ deux minutes en parallèle (≈2,9× plus rapides que les parcours Selenium équivalents), avec publication des résultats sur Microsoft Teams et suivi via un tableau de bord GitLab Pages.

Mots clés : Robot Framework, Browser Library, Playwright, tests smoke, tests end-to-end, CI/CD, GitLab, barrière qualité, flakiness, indépendance des tests, page object model, DevOps.

Design, Implementation, and Pipeline Integration of a Smoke Test Suite as a CI/CD Quality Gate for a Multi-Universe SaaS Platform

Abstract

LittleBig Connection runs a SaaS platform across four application universes. The QA regression suite (976 Robot Framework/SeleniumLibrary tests) took over eight hours with an unmeasured flakiness rate. This work introduces a smoke suite built with Robot Framework and Browser Library/Playwright that runs as the first GitLab CI/CD stage and blocks regression whenever a critical path fails. Eighteen independent tests cover the twenty highest-priority journeys across Auth, Admin, Client, and Supplier, complete in about two minutes in parallel (≈2.9× faster than equivalent Selenium paths), and publish results to Microsoft Teams with tracking through a GitLab Pages dashboard.

Keywords: Robot Framework, Browser Library, Playwright, smoke testing, end-to-end testing, CI/CD, GitLab, quality gate, flakiness, test independence, page object model, DevOps.

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